Contribution packet: regime selection, the format confound, and the timing constraint

For integration into the E1–E5 paper — A. Varaksin

PART 1 — drop-in text. 1433 words, about 4.3 pages in the Zapiski block (measured, not estimated). **That is more than the merged paper can afford** — Part 2 gives the arithmetic and a per-block cut order so you can take it down to whatever fits. **PART 2 (p. 5) — integration notes, not for the paper**; it also lists four edits the merge needs on the E1–E5 side.

PART 1: DROP-IN TEXT

Methods insert — add after "Metrics Implemented"

Rotation power. Winding is read off a per-trajectory 2D-PCA projection; we add a projection-free measure so the two cross-check. Taking the discrete Fourier transform along the unroll axis in the full 5280-dimensional state, *rotation power* R is the fraction of non-DC power in the period-6 bin over a 24-unroll tail — one scalar, no basis choice. It reproduces its own spectral definition to 4×10^{-8}; one threshold $R > 0.6677$ reproduces the hand-assigned rotate/settle label on **691 of 696** trajectories and **688 of 696** held out leave-one-bank-out; and rotating prompts peak at period 6 *specifically* (dominant bin exactly 6.00). Two cycles are needed, so 12 unrolls is the shortest window on which R is defined — a bound that matters in E9.

E6: Instruction wording, not the task, selects the regime

Holding the sequence, the required answer, the marker and the token count fixed at 53, only the noun in the instruction varies. "Report the largest **symbol**" rotates **36/36** paired cells; "largest **element**" and "largest **item**" rotate **0/36**. The selector is not semantic (neighbours of *symbol* rotate **0/144**, of *element* **96/120**), not orthographic (form-matched *symptom*, *cymbal*, *symmetry* rotate **72/72**), and not tokenisation (the 53-token group splits exactly 50%). Editing a single embedding row, prompt string and token identifiers untouched, flips the regime, both gates bit-exact at $0.000e{+}00$. Swapping the injected parameter e mid-trajectory, **62/64** switches take at points from unroll 1 to 48 — the regime follows the parameter the map currently has, not its history — entering rotation taking a median 15.5 unrolls against 1.0 to leave.

Scope. The within-regime control failed and norm-matching rescued only one noun pair of two, so the licensed statement is that the row determines the regime *within a small, direction-dependent neighbourhood*, not that the regime is a simple function of the embedding.

E7: The regime belongs to the prompt, not to the position measured

E1–E6 all read one token position, and the published figures for this model are drawn at *interior* ones, so we re-measured R at every position: rotating prompts rotate at **62.0%** of theirs (0.614–0.623, $n = 9$), settling prompts at **0.000**, no overlap. Onset is a single boundary — position 20 of 53 in *symbol*, where the second rule clause begins, not the selecting noun at 14. The answer-token restriction E1–E7 share therefore does not bias the labels; the check could have failed and did not.

E8: The accuracy measurements were measuring output format

Scoring the first generated token reads **0.125** where the answer is actually present in the generated text in **0.781** of cases, against a *measured* cross-item false-positive rate of 0.042–0.152. What displaces it is prose: *The* opens **39.4%** of 1260 banked generations. Depth makes the metric worse while the output improves — across $r = 2, 4, 8, 16, 32$ the rate of opening with *The* runs $0.032 \to 0.099 \to 0.631 \to 0.560 \to 0.647$ ($p = 1.09e{-}12$) while the rate of opening with the gold does not move ($p = 0.515$). Literally: `sub1` gold `3` $\to$ *"The answer is 4 - 1 = 3"*, scored wrong.

Supplying the format recovers the published benchmark number. Read from the released evaluation code, the ARC-Easy figure of 0.699 at $r = 32$ is produced zero-shot, normalised, with no option list; scored that way we obtain **0.658**, and five in-context examples under option-letter argmax independently give **0.723** (35 items fixed against 4 broken, $p = 3.35e{-}07$). Instruction fails where demonstration works: "reply with only the answer" gives no improvement, `\boxed{}` produced *fewer* such answers than not asking (2/36 against 4/36), and the system turn buys one item in twenty-four ($p = 1.0$), while two in-context examples move oracle accuracy by **+0.221** ($p = 9.3e{-}09$).

Conclusion: on this model a low exact-match score is not by itself evidence of a capability failure. Any accuracy claim needs its extraction rule stated and a measured chance rate beside it — over 1424 banked generations, exact match recovers 0.038 of the answers where a last-number rule recovers 0.240 at half the false-positive rate.

E9: The regime and the answer occupy non-overlapping windows

The rotation statistic is computed over the tail; the answer is best-ranked at a median unroll of about **four**. Measuring the same statistic on both windows separates two facts that had been conflated:

Window	Rotating	Settling	Separation
Decision (unrolls 0–11)	0.2292	0.2296	**0.0004**
Tail	0.862	0.298	**0.565**

A sliding 12-unroll window dates the divergence between window starts 12 and 16 (Figure 1). The early window is not quiet: its non-DC power exceeds the tail's by about two orders of magnitude (1.6×10^5 against 1943 and 59). It is energetic and simply carries no period-6 structure. So on this architecture the answer is fixed roughly twelve unrolls before the regime becomes measurable.

Scope — mandatory. This is a **timing claim, not a causal one**; it does not show that the regime cannot affect behaviour. It also bounds the instrument: R needs two cycles, so **12 unrolls is the earliest any period-6 statistic can speak at all**, and the non-separation is partly a property of the measurement rather than of the model. A behavioural null from manipulating the regime is correspondingly weak: a causally induced regime change altered the answer in 1 of 18 paired units, but acted on a property that did not yet exist when the readout settled.

Relation to E1–E5

The two sets of experiments use the same model, revision and hook, and they agree once the prompt sets are taken into account.

Why E1–E5 saw only settling. Every prompt in E1–E5 is written in one instruction style ("Question: … Answer:", "Start at 0. Add 1. Subtract 1. … "). E6 shows the regime is selected by instruction wording at fixed task, fixed answer and fixed token count. So "all 40 trajectories settle" is what E6 predicts for that wording, and E1–E5 become a large independent replication

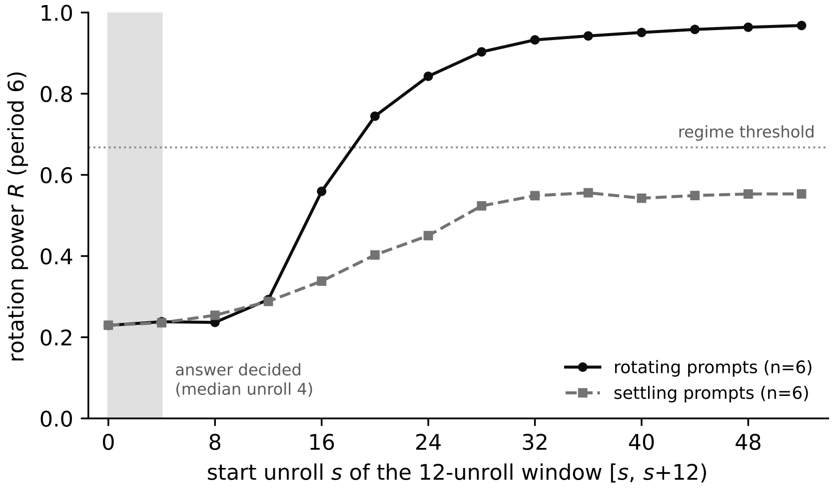

Figure 1: Rotation power in a sliding 12-unroll window, median over six rotating and six settling prompts. The axis is the window *start*: each point summarises unrolls $[s, s{+}12)$, twelve being the shortest window on which a period-6 orbit is defined. The families are indistinguishable until the window beginning at unroll 16, by which point the answer has long been settled (shaded; median unroll 4). Higher R means more power at period 6.

within the settling class — four task families, lengths 2–48, multiple seeds — rather than evidence that the model does not rotate. The results are complementary; neither is weakened by the other.

Two rotation measures, two scopes. Winding is a 2D-PCA quantity, R a full-space spectral one. E1's winding null and E6's positive result are measured on different prompt sets with different instruments and do not conflict. What the pairing shows is that a projection is fitted to the data it judges and confounds damping with rotation, so a winding null is a weak basis for a *negative* — a point E1–E5's own limitations already concede ("2D-PCA winding, sign-arbitrary"). Where a negative is asserted, R is the instrument that can carry it, having a threshold validated against held-out data.

H3. E3's length-matched control and the position taken here are compatible and sharper together. E3 shows effective computation grows with state retention and not with prompt length ($\rho = +0.558$ track against -0.576 local, both $p < 1e{-}8$). Independently, the running count *is* linearly decodable from the latents, and an untrained model carries the lagged count at cross-validated $R^2 = 0.7498$ against the trained model's 0.7175. Joint reading: the model spends more compute on state-retention tasks *and* the state survives contraction. What fails is H3's strong form — "contraction implies the model cannot maintain a running count" — not E3, which is untouched. Relatedly ρ should not be quoted as a single model constant: it is family-level, spanning 0.8239 to 0.9181, with between-family spread about four times the within-family spread.

Limitations — add to the existing list

- These accuracy figures and the published one compared against share an uncorrected surface-form bias inherent to likelihood scoring; the comparison survives it, the absolute values of both are questionable in the same direction.
- E9 says *when* the properties become measurable, not *whether* one influences the other; an intervention inside the decision window could still find an effect.
- A direct behavioural test of H3, injecting a donor state at h_0, failed three times for harness-internal reasons and is not reported; the authors' reported path independence over h_0 partly predicts a null there.
- Steering e with a difference-of-means vector moved neither behaviour nor geometry ($|\Delta R| \leq$

0.0065, zero crossings in 132 conditions, identity gate exact). The dose was adequate, so we read it as the wrong *injection site* — onset is 33 positions upstream — not as orthogonality of directions.

Contribution — add a column to the matrix

Arsenii Varaksin. Rotation-power statistic and its validation; regime experiments E6 and E7; the format-versus-capability decomposition E8; the window split E9; the steering experiment; and the instrument-null and retraction record.

PART 2: INTEGRATION NOTES — NOT FOR THE PAPER

The page budget, measured

Stripping markup and counting body prose only, against the Zapiski text block at 330 words/page (calibrated on a known 10-page build of the same class):

	Words	Pages (prose only)
E1–E5 body	1546	≈ 4.7
This packet, Part 1	1433	≈ 4.3
Together	2979	≈ 9.0

Nine pages of prose, before the abstract and title block (≈ 0.7), the eight tables in E1–E5 and the two here (0.2–0.3 each, so 2.0–2.5), the figure (≈ 0.35) and the bibliography (≈ 0.7). Realistically the merged body lands near **12–13 pages against a limit of 10**, so roughly 700–900 words have to come out of one side or the other. I am not in a position to judge which of E1–E5 is expendable, so here is mine costed and ranked instead — cut from the bottom up.

Block	W	Pp	Priority
Relation to E1–E5	336	1.0	**Essential.** Without it the merged paper contradicts itself in three places.
E8 (format)	249	0.8	**Essential.** Bears directly on the 0% counting claim (note 2).
E6 (regime)	173	0.5	**Essential.** The headline result.
E9 (timing)	288	0.9	Important. The hinge. Text must stay if the claim is made at all; the figure can go (-0.35 pp).
Methods insert	123	0.4	Important. This is the winding/R interface; without it E6 and E1 look like a contradiction.
Limitations	143	0.4	Keep the first two bullets if pressed.
E7 (position)	85	0.3	**Cut first.** A validity check, not a result.
Contribution	36	0.1	Required by the venue.

Dropping E7, two limitations bullets and the figure takes this to ≈ 3.3 pages without losing a claim. Below that, claims start going, and I would rather you tell me which than guess. The full-length treatment of all of it is in the repository (`docs/submission/build/main.pdf`, 10-page body), so nothing is lost from the record by cutting here.

Four edits the merge needs on the E1–E5 side

The first two are self-contradictions a reader would find inside a single document; the third is a load-bearing evidence gap; the fourth is a scope question.

1. "H2 is refuted" has to become narrower — highest priority. E1 and the conclusion read "naive H2 is refuted" and "no winding-depth signal detected (naive H2 refuted)". H2 as the adaptive-computation literature means it is *per-position* depth allocation, and Huginn has no halting head: `num_steps` unrolls the whole sequence together, every position receives exactly r, and there is nothing to allocate. The hypothesis is not expressible on this architecture, so no null can refute it. What E1 genuinely refutes is the *winding operationalisation* — that winding number grows with required reasoning steps. Suggested wording: *"the winding operationalisation of H2 is refuted on this prompt set; H2 in the ACT/Universal-Transformer/PonderNet sense is* *not expressible on an architecture without a halting mechanism, so no experiment here bears on it.”*

2. “The model fails synthetic counting (0% at len $\geq$ 8)” needs its scoring rule stated. This is the claim E8 most directly threatens, and it is load-bearing for Discussion point 2 (“counting failure is theoretically predicted”). On our census families exact-match scoring read 0.000 where the answer was present in the generated text in 78.1% of cases; a 0% reading on this model is exactly what a formatting confound produces. If the counting accuracy came from exact match or a first-token/fixed-parser rule, the banked generations should be re-scored by containment before the capability claim stands. It may well survive — parity and state tracking are predicted to fail on theoretical grounds — but the number needs the format control, or the claim should read “fails under exact-match scoring” with the rule named.

3. The Pappone refutation should be split by which arm supports it. Two different experiments currently sit under one heading: a *synthetic* arm (depths 2–5, orthogonality ≈ -0.5, exits fixed at 50%) run on a purpose-built 128-dimensional recurrent autoencoder trained for 50 epochs, and a *FineWeb* arm on real text (cos ≈ -0.687). Only the second can speak to Pappone et al., whose claim concerns recurrent-depth transformers. The toy model also *initialises* its rotation layer orthogonally (`nn.init.orthogonal_`), so orthogonal structure there is partly imposed by construction rather than discovered — worth one clause, because a reviewer who opens the notebook will see it. Suggested split: keep “step normalisation is required for a clean orthogonality measurement” as the methodological finding, which both arms support and which is the genuinely useful result; restrict “refutes the universality of Pappone et al.” to the FineWeb arm.

A correction I owe on this: I earlier flagged the README’s −0.687 as inconsistent with the −0.494 in `two_scale_real.csv`. That was wrong. They are two different experiments — FineWeb real text and synthetic depths 2–5 respectively — and both tables in the paper are internally consistent. The filename misled me.

4. Duplicate material to delete on merge. Both drafts carry a “Model and Protocol” paragraph naming the same revision (`bb6621b6...`) and the same `core_block[-1]` hook; keep one. Four bibliography entries are shared and will duplicate: Geiping, Bai (DEQ), Grazzi, Pappone. The E1–E5 draft cites Merrill (2404.08819) for “illusion of state” and Grazzi (2411.12537) for parity, which is correct — do not let a merge collapse them, since an earlier draft of ours cited 2404.08819 for the parity claim and that was a misattribution we had to fix.

Not a conflict, worth one sentence. The limitation “Geiping regime not reproduced — we did not use their system prompt or $r = 128$” now has a partial answer: the system turn is not the missing ingredient. The identical instruction text in the system turn rather than the user turn buys one item in twenty-four, paired, $p = 1.0$.

On the compliance items, since the merge changes who owns them

The venue requires a Contribution section with one entry per team member (stated twice in the instructions), all authors registered on OpenReview, and mentors listed. The E1–E5 contribution matrix currently has columns only for Shiianov and Sverdlov; it needs a third. The acknowledgments there thank S. Barannikov and N. Kurdyukov, while our draft lists Barannikov as mentor — worth agreeing which is correct before upload, as it is a stated requirement rather than a matter of taste.